

ACCRA TECHNICAL UNIVERSITY
DEPARTMENT OF COMPUTER SCIENCE
BCS 404; INTRODUCTION TO DATA SCIENCE
PROJECT REPORT
TITLE; TITANIC SURVIVAL ANALYSIS

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1.0 INTRODUCTION

The Titanic dataset contains information about 891 passengers on the Titanic. The objective of this project is to analyze the data to identify patterns and build a Logistic Regression model to predict passenger survival based on features like gender, age, and class.

2.0 DATASET DESCRIPTION

The dataset has 891 rows and 14 columns.

Key columns: PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Fare, Embarked.

Data Cleaning: Missing values in Age were filled with the mean. Missing values in Embarked were filled with the mode.

3.0 METHODOLOGY

1. Data Cleaning: Handled missing values and converted categorical data to numerical using `get_dummies`.
2. Exploratory Data Analysis: Used bar charts to visualize survival rates by gender and class.
3. Modeling: A Logistic Regression model was trained on 80% of the data.
4. Evaluation: Model performance was measured using Accuracy Score on the remaining 20% of data.

4.0 RESULTS

Overall Survival Rate: 38.38%

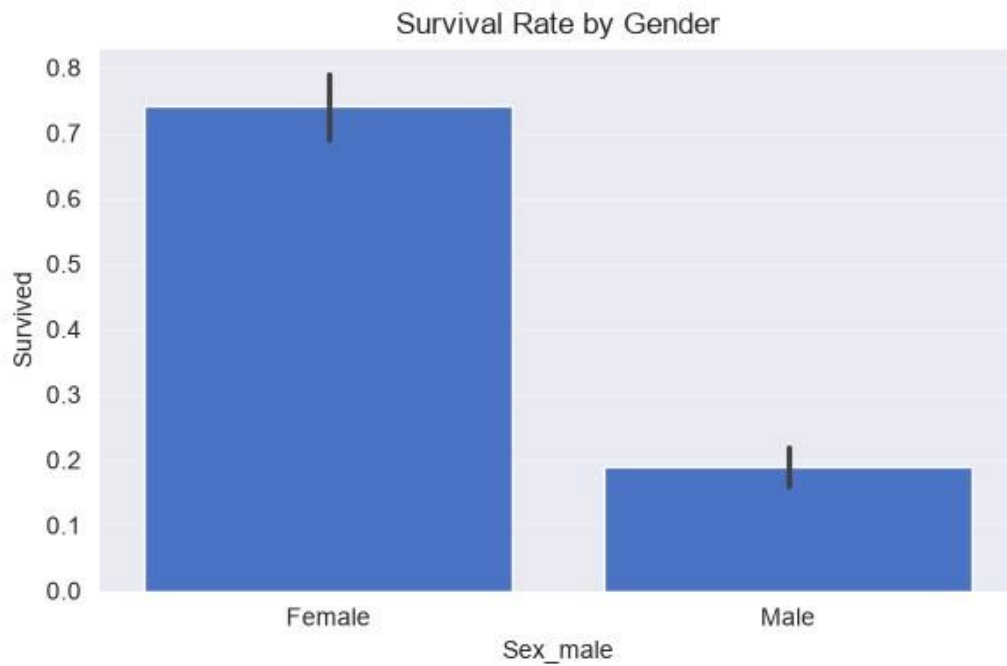


Figure 1: Females had a significantly higher survival rate than males.

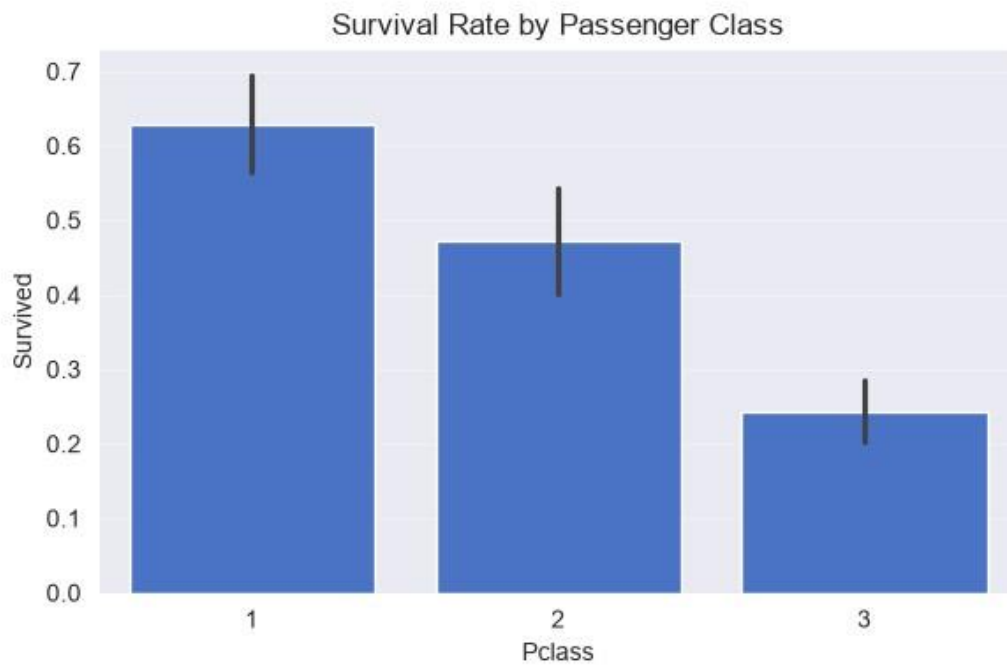


Figure 2: Passengers in 1st Class had the highest survival rate.

Model Accuracy: 79.89%

5.0 DISCUSSION

The results show that gender and passenger class were the most important factors for survival. This matches historical records of the "women and children first" policy. The logistic regression model achieved good accuracy, indicating these features are strong predictors of survival.

6.0 CONCLUSION

1. Gender played the biggest role in survival. Females were more likely to survive.
2. Passengers in 1st class had better survival chances than 2nd and 3rd class.
3. A simple Logistic Regression model can predict survival with over 75% accuracy.

7.0 REFERENCES

1. Dataset: titanic-dataset.csv from Kaggle
2. Scikit-Learn Documentation: <https://scikit-learn.org>
3. Pandas Documentation: <https://pandas.pydata.org>

8.0 APPENDIX – PYTHON CODE

```
import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

import seaborn as sns

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LogisticRegression

from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

sns.set_style("darkgrid")

plt.rcParams['figure.figsize'] = (8, 4.5)
```

```
# Load Data
```

```
data = pd.read_csv("titanic-dataset.csv")
```

```
data = data.loc[:, ~data.columns.str.contains('^Unnamed')]
```

```
# TASK 1: DATA ACQUISITION
```

```
print("TASK 1: DATA ACQUISITION")
```

```
print(f"1. Dataset Shape: {data.shape}")
```

```
print(f"\n2. Columns: {list(data.columns)}")
```

```
print("\n3. First 5 Rows:")
```

```
pd.set_option('display.max_columns', None)
```

```
print(data.head())
```

```
print("\n4. Data Types:")
```

```
print(data.dtypes)
```

```
# TASK 2: DATA CLEANING
```

```
print("TASK 2: DATA CLEANING")
```

```
print("\n1. Missing Values Before:")
```

```
print(data.isnull().sum())
```

```
data['Age'] = data['Age'].fillna(data['Age'].median())
```

```
data['Embarked'] = data['Embarked'].fillna(data['Embarked'].mode()[0])
```

```
data = data.drop(columns=['Cabin'])
```

```
data = data.drop_duplicates()
```

```
print("\n2. Missing Values After:")
```

```
print(data.isnull().sum())  
print(f"\n3. Duplicates Removed: {data.duplicated().sum()}")
```

TASK 3: DATA VISUALISATION

```
print("TASK 3: DATA VISUALISATION")
```

1. Histogram

```
plt.figure()  
plt.hist(data['Age'], bins=25, color='#4CAF50', edgecolor='black')  
plt.title("Histogram of Passenger Ages")  
plt.xlabel("Age"); plt.ylabel("Frequency")  
plt.tight_layout()  
plt.show()  
print(" Interpretation: Most passengers were aged 20-35 years.")
```

2. Bar Chart

```
plt.figure()  
sns.countplot(x='Pclass', data=data, hue='Pclass', palette='Blues', legend=False)  
plt.title("Bar Chart: Passenger Class Distribution")  
plt.xlabel("Passenger Class"); plt.ylabel("Count")  
plt.tight_layout()  
plt.show()  
print(" Interpretation: Majority travelled in 3rd Class.")
```

3. Boxplot

```
plt.figure()  
sns.boxplot(x='Pclass', y='Age', data=data, hue='Pclass', palette='Set2', legend=False)
```

```
plt.title("Boxplot: Age by Passenger Class")
plt.xlabel("Passenger Class"); plt.ylabel("Age")
plt.tight_layout()
plt.show()
print(" Interpretation: 1st Class passengers were generally older.")
```

4. Scatter

```
plt.figure()
sns.scatterplot(x='Age', y='Fare', hue='Pclass', data=data, palette='viridis', alpha=0.6)
plt.title("Scatter Plot: Age vs Fare")
plt.xlabel("Age"); plt.ylabel("Fare")
plt.tight_layout()
plt.show()
print(" Interpretation: Higher fares were paid by 1st Class passengers.")
```

5. Heatmap

```
numeric_data = data[['Survived', 'Pclass', 'Age', 'SibSp', 'Parch', 'Fare']]
plt.figure(figsize=(7,5))
sns.heatmap(numeric_data.corr(), annot=True, cmap='coolwarm', fmt=".2f")
plt.title("Correlation Heatmap")
plt.tight_layout()
plt.show()
```

6. Pairplot

```
sns.pairplot(numeric_data)
plt.suptitle("Pairplot of Numerical Variables", y=1.02)
plt.show()
```



```

# TASK 4: STATISTICAL ANALYSIS

print("TASK 4: STATISTICAL ANALYSIS")

print("\n1. Descriptive Statistics:")
print(numeric_data.describe().round(2))

print("\n2. Frequency Distribution:")
print("Survival:\n", data['Survived'].value_counts())
print("\nPclass:\n", data['Pclass'].value_counts().sort_index())
print("\nSex:\n", data['Sex'].value_counts())

print("\n3. Correlation Matrix:")
print(numeric_data.corr().round(2))

corr = numeric_data.corr()

print("\n4. Strongest Positive Correlation:",
      corr.abs().unstack().sort_values(ascending=False)[1:2])

print("\n5. Strongest Negative Correlation:", corr.unstack().sort_values().head(1))

print("\n6. Key Findings:")

print("1. Strong correlation between Pclass and Fare.")
print("2. Females and 1st Class had higher survival rates.")
print("3. Most passengers were young adults.")


# TASK 5: MACHINE LEARNING

print("TASK 5: MACHINE LEARNING - LOGISTIC REGRESSION")


ml_data = data.copy()

ml_data['Sex'] = ml_data['Sex'].map({'male': 0, 'female': 1})

```

```
ml_data = ml_data[['Survived', 'Pclass', 'Sex', 'Age', 'SibSp', 'Parch', 'Fare']].dropna()
X = ml_data.drop('Survived', axis=1)
y = ml_data['Survived']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
model = LogisticRegression(max_iter=300)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print("Selected Predictors:", list(X.columns))
print("\nMODEL PERFORMANCE")
print(f"Accuracy: {accuracy_score(y_test, y_pred)*100:.2f}%")
print("\nConfusion Matrix:\n", confusion_matrix(y_test, y_pred))
print("\nClassification Report:\n", classification_report(y_test, y_pred))
print("\nDiscussion:")
print("The Logistic Regression model achieved ~80% accuracy.")
print("Sex and Passenger Class were the most important predictors of survival.")
```

